JAWAHAR RANGANATHAN

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EDUCATION

Birla Institute of Technology and Sciences KK Birla Goa Campus 2022 – Present | Goa
B.E. (Hons.), Computer Science and M.Sc. (Hons.), Economics
CGPA: 7.3

GD Goenka Public School, *CBSE* 2009 – 2022 | Delhi
Class X - 92.4%
Class XII - 93.6%

TECHNICAL PROFICIENCY

C, C++, C#, Python, Java, SQL , R , JavaScript , Verilog, AWS, Linux , Wireshark, Git

SUBJECTS / ELECTIVES

Machine Learning, Digital Design, Foundations of Data Science, Data Structures & Algorithms, System Design, Operating Systems, Object Oriented Programming, Data Science, Database Systems

PROJECTS

Altruistic Ride-Sharing - Multi-Agent Optimization - Reinforcement Learning 10/2025 – Present

- Built a **multi-objective ride-sharing framework** using multi-agent reinforcement learning (MADDPG) with altruism based incentives for scalable, real-world urban mobility systems to optimize fleet efficiency across complex, high-demand environments
- **Introduced an Altruism Score** to model driver-rider reciprocity, enforce fairness, and stabilize non-stationary learning under stochastic population turnover and participation by incentivizing cooperative behaviors that maximize collective network utility
- **Validated via A/B simulations** against profit-driven baselines, achieving **higher utilization, lower detours and CO2 emissions**, and stable long term system equilibria while outperforming traditional greedy algorithms in volatile market conditions.

Lesion Segmentation in Medical Imaging - Deep Learning 10/2025 – Present

- Built an **end-to-end supervised lesion segmentation pipeline** with a forensic audit to infer dimensionality, mask structure, and verify NPZ-DICOM alignment and leakage free splits securing data integrity across complex and sensitive medical datasets.
- Implemented **data handling and preprocessing (automatic NPZ key inference, strict image-mask alignment, empty-mask negatives, CT windowing)** with reproducible training minimizing bias while streamlining large-scale clinical deployment
- Evaluated using **Dice/IoU and sensitivity/precision,** with qualitative overlays to assess clinical plausibility and failure modes **delivering rigorous performance benchmarks** and **actionable clinical insights**

Protein Motif Detection for Vaccine Target Identification – Genomics 05/2025 – 11/2025

- Built a **hybrid sequence structure motif mining pipeline** using **attention-based embeddings** and **epitope prediction** to identify conserved surface exposed viral regions facilitating the discovery of universal targets for pan-virus vaccines
- **Analyzed 1,200+** viral proteomes, **achieving >87% cross-strain conservation** while **reducing the vaccine target space by 65%** and preserving neutralizing epitopes to dramatically accelerate laboratory validation and clinical trials
- Enabled **rapid antigen prioritization** for emerging variants, supporting peptide vaccine design in outbreak and low-resource public health settings by **providing a cost-effective framework** for urgent immunization strategies

INTERNSHIPS

Research Intern, *Temple* 06/2026 – Present | Gurgaon

- Developed **deep learning workflows** for wearable sensor data to support research team experimentation and model evaluation
- Built **neural network pipelines** for wearable signal analysis, including **preprocessing, feature learning, and performance validation**
- Worked with TEMPLE research team on **wearable intelligence** models for robust **real-world physiological signal analysis**

Data Analytics Intern, *Roche* 06/2025 – 07/2025

- Modeled and segmented large-scale oncology HCP datasets using **AWS Athena** to derive analytical features
- Analyzed therapy adoption and multichannel engagement patterns for **data-driven HCP profiling**
- **Built scalable SQL analytics pipelines and dashboards** to support precision targeting and ML-ready datasets

POSITION OF RESPONSIBILITY

International Conference on Cognitive Neuroscience, *Chief Organizer* 04/2024 – 04/2025

- Oversaw **end-to-end conference planning, stakeholder coordination, and program execution** across two international events
- **Led global delegate logistics** and managed on-ground execution for international researchers and invited speakers

Birla Institute of Technology and Sciences, *Teaching Assistant* 02/2025 – 05/2025

- Mentored **300 undergraduates** in combinatorics and probability, building rigorous university-level quantitative reasoning
- Conducted **tutorials and evaluations** to ensure mastery of discrete mathematics, logic, and probability

AWARDS AND RECOGNITIONS

Junior Science Talent Search Examination Rank Holder, *Science Branch of the Directorate of Education, Delhi* 01/2019

- **Achieved State Rank 85 in JSTSE 2019, outperforming 30,000 students** in a rigorous assessment of scientific and analytical aptitude.

National Standard Examination in Junior Science Merit List, *Indian Association of Physics Teachers (IAPT)* 11/2018

- Qualified for the NSEJS Merit Award Scheme, placing within the **top 300 out of 80,000+** national candidates.